

Davinder Singh

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Summary — Postgraduate student with a background in computer science and a developing focus in quantum machine learning. Currently working on designing a fully quantum LSTM architecture as part of master's thesis, exploring the use of entanglement and partial measurements for recurrent connections. Interested in the intersection of quantum computing and machine learning, with a long-term goal of contributing to research in quantum models for learning and cognition. Demonstrates strong self-motivation, independent problem-solving skills, and a commitment to academic and research development.

Skills

Language Python, C++, JavaScript

Quantum PennyLane, Qiskit, Cirq

ML & DL PyTorch, TensorFlow, NumPy, SciPy, scikit-learn

QML VQC, PQC, Hybrid Models, Partial Measurements

Tools Git, Jupyter, VS Code, Bash

Experience

M.Tech Thesis – Quantum Machine Learning

Aug 2023 – Present

Advisor: Dr. Debajyoti Bera, IIIT-Delhi

Project: Fully Quantum LSTM Architecture

- Designed and implemented a novel fully quantum recurrent neural network using entanglement and partial measurements to emulate memory dynamics.
- Built modular, customizable architecture in PennyLane, integrating parameterized quantum circuits (PQCs) for learnable transformations.
- Developed hybrid training strategies and evaluated expressivity compared to classical and hybrid models.
- Preparing results and implementation for preprint publication by June 2025.

Education

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)

Master of Technology in Computer Science

July 2023 – July 2025

Specialisation: Artificial Intelligence

Amritsar College of Engineering and Technology

Bachelor of Technology in Computer Science

Aug 2015 – Aug 2019

Relevant coursework in general Computer Science and Engineering

Projects

Multi-Lingual Complex NER

Feb 2024 – May 2024

- Built an ensemble of multilingual RoBERTa models (Hindi, English, Bengali) for named entity recognition.
- Trained models to accurately extract complex named entities from text data.
- Completed as a group project for the Natural Language Processing course at IIIT-Delhi.

Quantum NLP Library (QNLP)

Jan 2024 – Present

- Designing a lightweight Python library to experiment with Quantum Natural Language Processing using hybrid models.
- Implemented modules for embedding textual data into quantum states and building variational circuits for NLP tasks.
- Integrated PennyLane backends and provided interfaces for custom data loaders, quantum layers, and evaluation pipelines.
- Aiming to provide educational clarity and prototyping ease for research in QNLP.

Achievements

- Secured **AIR 905 in GATE Computer Science 2023** with a score of 634.
- Awarded the **Qiskit Developer Certificate of Excellence** during Qiskit Global Summer School 2024.
- Winner of the **Qiskit Quantum Challenge 2024**, showcasing problem-solving skills in quantum circuit design and algorithm optimization.